

COLOSSUS 3.0



Team Name: Worriors

Project / Idea Title: First Response System

Track: healthtech

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The Problem - Why Does It Matter?

01 PROBLEM STATEMENT

Every second counts in a medical emergency, yet victims cannot share live location, reach the right hospital, or summon help fast enough. Responders act blind — without a shared, real-time picture of the crisis.

02 WHO IS AFFECTED?

Road-accident victims, cardiac & stroke patients, and the elderly living alone — plus families left in the dark, ambulance crews, under-prepared emergency rooms, and police struggling to clear routes.

03 CURRENT PAIN POINTS

108/112 voice calls with no live GPS; hospitals admit “blind” to incoming critical cases; no ambulance tracking; no link between hospitals and traffic police; families notified far too late.

Idea / Solution

01 WHAT WE'RE BUILDING

One coordinated platform with four live dashboards — Patient (panic button + auto crash detection), Ambulance (GPS dispatch & journey stages), Hospital (ER readiness & equipment), Police (green corridors) — on a shared real-time backend.

02 HOW IT SOLVES IT

Panic or crash instantly triggers GPS + SMS alerts, auto-assigns the nearest suitable ambulance (ICU-aware), draws live OSRM routes, prepares the ER, and lets police block road segments that force instant re-routing.

03 CORE VALUE PROPOSTION

One tap to reach help. We compress the golden hour — location, dispatch, route, corridor and hospital readiness handled in a single automated loop.

How It Works

01 END-TO-END WORKFLOW

Crash / panic → 10-second cancel countdown → incident created with GPS → nearest ambulance auto-assigned → live route drawn → police clear corridor → ER prepared → handoff at arrival.

02 USER JOURNEY

Patient: register (Aadhaar) → grant location → panic. Ambulance: accept → navigate → onboard → deliver. Hospital: incoming queue + equipment. Police: draw road-block → system re-routes.

03 TECHNOLOGY STACK

Node.js · Express · Socket.IO · Leaflet + OpenStreetMap · OSRM routing · Web Geolocation & DeviceMotion · Fast2SMS / Twilio · JSON persistence · Cloudflare Tunnel.

What Makes Us Different?

01 CORE FEATURES

4-role dashboards · auto crash detection with 10s cancel · ICU-aware dispatch · live ambulance & hospital routes · road-block re-routing · ER equipment board · SMS + WhatsApp alerts.

02 EXISTING APPROACH

Voice calls to 108/112, manual bed-availability phone calls, standalone ambulance or hospital apps with no shared state, and police coordination by radio — slow, blind, fragmented.

03 OUR INNOVATION / USP

DeviceMotion auto-SOS that detects a crash without user action; police road-segment blocks that automatically reassign blocked routes; ICU-capability matching — one live coordination loop for all four stakeholders.

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Impact & Execution

01 EXPECTED IMPACT

Faster dispatch and earlier ER preparation aim to cut emergency response time by 30–40%, directly improving survival odds inside the golden hour.

02 BENEFICIARIES

Patients & families · ambulance services · hospitals and ERs · traffic police · city emergency management bodies.

03 SCALABILITY

City → state → national rollout; pluggable SMS providers; JSON store upgradeable to PostgreSQL; containerized deployment on any cloud.

04 IMPLEMENTATION ROADMAP

Phase 1: core platform (done) → Phase 2: real SMS + push notifications → Phase 3: hospital & traffic-signal integrations → Phase 4: AI dispatch + multi-city rollout.

Thank You